I/O Workload Analysis of AI Applications for Autonomous Vehicles with Shared Storage

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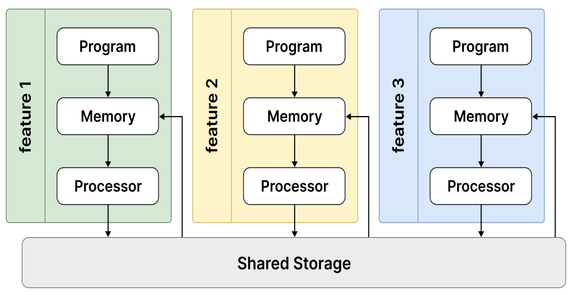
Introduction

Why Use Shared Storage?

Autonomous vehicles require real-time processing of vast sensor data [1]. High-performance shared systems are used for efficient storage and access.



Why Analyze I/O Patterns?

Analyzing I/O patterns, influenced by traffic conditions, is crucial for optimizing storage system performance [2]. It helps eliminate data processing bottlenecks, enhance system reliability by detecting abnormal behaviors, and improve storage efficiency.

How Was The Analysis Conducted?

We operated two AI models for 3600 seconds and recorded the I/O patterns using blktrace.

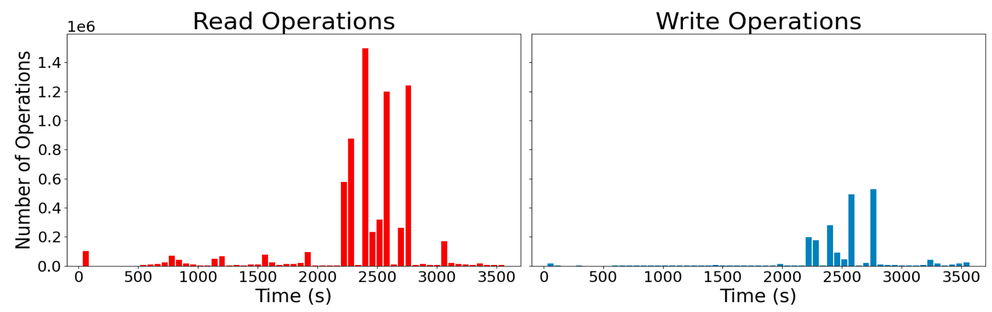
1. We compared the number of read and write operations.
2. We used the standard deviation of latency as a measure to determine latency variability and examined the latency variability for both Q2D (Queue to Device) and D2C (Device to Complete) [3].

Experiment 1 : SAM

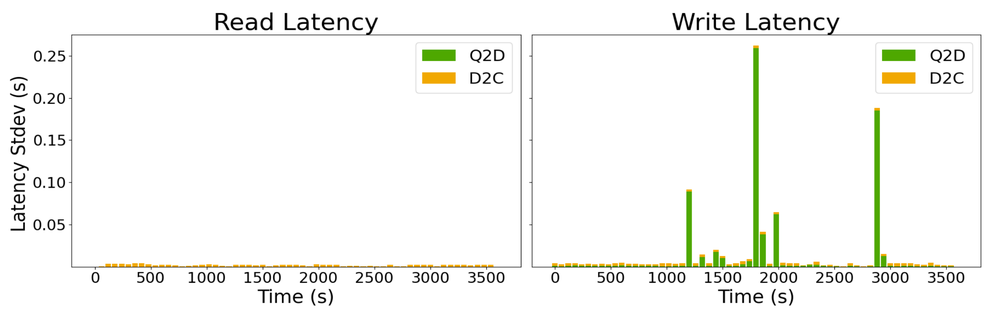
Model : SAM (Segment Anything Model)

- SAM is a model that identifies and classifies various objects simultaneously.
- In autonomous vehicles, it can be used to detect and analyze traffic signs, pedestrians, vehicles, and more in real-time.

Analysis Result



1. In the process of segmenting image data, the SAM performs numerous read operations, while write operations occurs after a certain period.



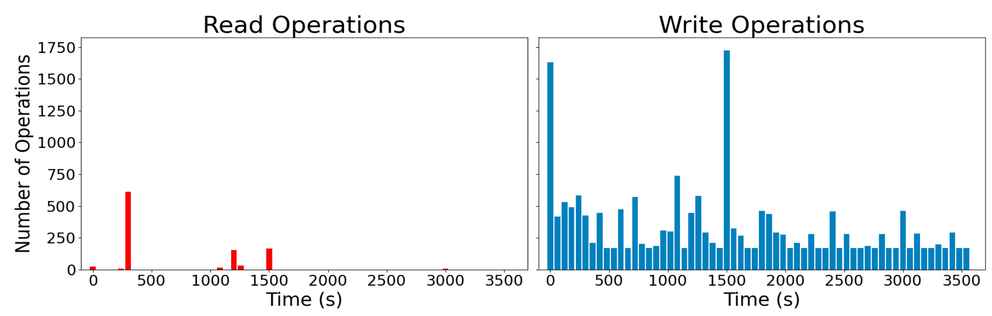
2. The values for Q2D were relatively high, suggesting that there may be variability in latency at the operating system level during write operations. The values for D2C were consistently low. Indicating stable performance.

Experiment 2 : OpenCOOD + V2VNet

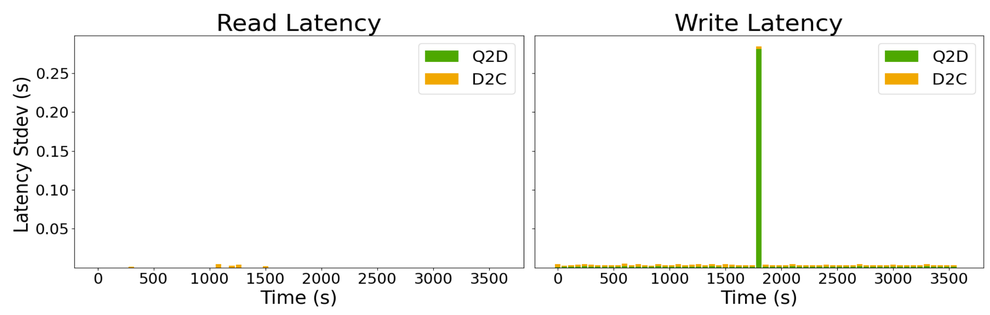
Model : OpenCOOD + V2VNet

- OpenCOOD is software that simulates data sharing and collaborative object detection through V2V, V2I communication.
- V2VNet is a model that enhances autonomous driving by sharing and integrating information between vehicles in real-time.

Analysis Result



1. We observed a high number of write operations, which is because the large amount of information exchanged between vehicles contributes to the significant amount of data that needs to be recorded.



2. The overall standard deviation in latency is very low. However, there are periods where the standard deviation of Q2D during write operations increase. This seems to be influenced by external factors or system effects.

Conclusion

1. Since the frequency of each task in the model varies, it is necessary to determine which tasks should be prioritized for optimization
2. In actual autonomous vehicles, multiple models operate simultaneously, so the shared storage system must be designed to ensure stable I/O latency
3. This study measured under limited AI model operating conditions. Future research with various operating conditions will help improve storage performance

Reference

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3. H. Liu, Z. Wang, and J. Zhao, “Cola: Characterizing and optimizing the tail latency for safe level-4 autonomous vehicle systems,” arXiv preprint arXiv:2305.07147, 2023